

Blood parasite infection is associated with lower thermoregulatory precision in the lizard *Podarcis lusitanicus*

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ABSTRACT

Parasites have the potential to disrupt and induce changes in thermoregulation and behaviour in ectotherm hosts, possibly with deleterious effects. In a previous study, the lizard *Podarcis lusitanicus* (Lacertidae), when infected with blood parasites of the genus *Karyolysus* (suborder Adeleorina), were bolder, remained exposed for longer, and tended to explore more. To determine if this behaviour was linked to the lizards' thermoregulatory behaviour, we recorded the temperatures of male lizards with varying blood parasite infection levels in a thermal gradient during their active diurnal period. Infected lizards had lower thermoregulatory precision, indicating an association with the previously observed higher exploratory behaviour of infected individuals. However, parasite infection was not significantly associated with higher median or maximum recorded body temperatures, suggesting that the lizards were not compensating through behavioural fever. While we observed no significant effect on leucocyte counts from the parasite infection or thermal variables, the latter had a different magnitude of effect on the host cellular immune response depending on whether the lizard was parasitised or not. The cellular immune response of parasitised lizards increased less with increasing thermal variables (preferred body temperature, maximum recorded temperature, and thermoregulatory precision), suggesting the parasite infection might be negatively affecting the development of stronger immune responses in these hosts. It is also possible that changes to host thermoregulation and behaviour might benefit the parasite beyond its infectious stage in the intermediate vertebrate host, because increasing exploratory behaviour in the lizard could increase the probability of contact with haematophagous mites, the definitive hosts of *Karyolysus* sp. Finally, while tail condition was not significantly related to thermal preference, lizards with regenerated tails had significantly higher thermoregulation precision than lizards with original tails, highlighting the need for additional studies on the real impact of tail autotomy and regeneration in thermoregulatory capabilities of lizards.

1. Introduction

Thermoregulation is fundamental to ectothermy, as body temperature directly impacts physiological and biochemical processes, growth and development, life history, behaviour and activity (Adolph and Porter, 1993; Buckley, 2008; Woods et al., 2003). Therefore, anything affecting the underlying processes of thermoregulation could have significant impact on an organism's condition and physiological performance (Black et al., 2019). While climate change and habitat destruction

are some of the most widely cited causes for disrupting thermoregulation patterns in ectotherms (Deutsch et al., 2008; Sinervo et al., 2010), another often overlooked candidate is parasitism. Across ectotherm hosts, ecto- and endoparasites can disrupt their host's thermoregulation in various ways, both directly and indirectly (e.g.: Elliot et al., 2002 for insects; Goodman and Johnson, 2011 for amphibians; Macnab and Barber, 2011 for fish). For lizard hosts, current evidence regarding how parasites impact host thermoregulation is mixed (Johnson et al., 2019; Scholnick et al., 2010). The type of parasite may play a role, for

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example, intestinal coccidia (endoparasites) disrupt thermoregulation-related responses to tick infestation (ectoparasites) in lizards (Megía-Palma et al., 2020a). In terms of host response, lizards themselves may modulate thermoregulation in response to parasitic infection by seeking greater thermal exposure in an attempt to raise their body temperature or by increasing basking time (Scholnick et al., 2010; Vaughn et al., 1974). This behaviour, referred to as behavioural fever, is thought to mitigate the effects of infection (Rakus et al., 2017). Behavioural fever has been observed across vertebrate ectotherms in response to a variety of pathogens and parasites (e.g. Huntingford et al., 2020; Nasri et al., 2022; Ryan et al., 2018; Sauer et al., 2019). Conversely, there are several examples in the literature of behavioural hypothermia in ectotherms infected with parasites, where hosts reduce their activity and body temperature, thereby making themselves less suitable for parasite development (Lefcort and Bayne, 1991; Moore and Freehling, 2002; Oppliger et al., 1996; Paranjpe et al., 2014). For lizard hosts, this may be accompanied by changes in thermoregulatory precision (Horváth et al., 2020; Maia-Carneiro et al., 2018; Megía-Palma et al., 2020b). However, the directionality of the effects in this trait are less clear, as both decreases (Horváth et al., 2020) and increases (Megía-Palma et al., 2020b) in thermoregulatory precision have been identified in parasitised lizards.

At the same time, some studies report only a negligible effect of parasitism on lizard hosts' fitness-related traits, and aspects such as host's mobility and body condition, which are two commonly used proxies to evaluate lizard health (Bonneaud et al., 2017; Damas-Moreira et al., 2022; Lee et al., 2023; Oppliger et al., 2004; Schall, 1983; Spence et al., 2017, but also see Oppliger et al., 1996). However, a study that included data for 10 species of lacertid lizards found sex-specific effects on two indices of host body condition from infection by one of the most common genera of blood parasites found in western Mediterranean lizards (Megía-Palma et al., 2024a). The apparent lack of a negative effect of blood parasites in previous studies (Bonneaud et al., 2017; Damas-Moreira et al., 2022; Schall, 1983) may be explained by both evolutionary causes as some host-parasite co-evolutionary histories can promote a level of tolerance to infections in hosts (Faria et al., 2024; Megía-Palma et al., 2023; Schall, 1990). Lizard host immunocompetence, especially mechanisms arising from these co-evolutionary processes, is also likely to be involved in these relationships, as it is both linked to the presence of pathogens, including parasites, and to thermoregulatory constraints (Duran et al., 2020; Madelaire et al., 2021; Zamora-Camacho et al., 2016). Cellular immune response, mediated by the presence and differentiation of leucocytes in the blood, is an immunocompetence parameter commonly considered for lizards in parasitism studies (Damas-Moreira et al., 2022; Madelaire et al., 2021), which usually optimally operates at stable temperatures near the lizard's preferred body temperature (Duran et al., 2023; Zimmerman et al., 2010). Results from non-lacertid lizards hosting blood parasites vary across host-parasite species pairs and locations, with some finding a significant reduction of cellular immune response in parasitised individuals in one region but not for the same host-parasite pair in a neighbouring region (Bessa et al., 2020). Contrarily, studies in other lizard species have found evidence of immune cellular response being stimulated by parasite infection (Motz et al., 2014), and, in yet another species, evidence of increased immune competence but not associated with cellular immune response (Spence et al., 2017).

In a previous study, blood parasites (Adeleorina, Apicomplexa) affected the exploratory behaviour of the lizard *Podarcis lusitanicus* (Squamata: Lacertidae) in a controlled environment, with parasitised individuals more willing to leave a refuge and spend more time basking under a light source (Faria et al., 2024). Here, we test the hypothesis that this behaviour is associated with changes in thermoregulation because of infection. Considering the increased tendency towards exploratory activity found in parasitised *P. lusitanicus* (Faria et al., 2024), we predicted that parasite infection negatively impacts the host's thermoregulatory precision, in turn causing parasitised lizards to move more, and

thereby appear bolder, since they may be unable to precisely maintain their preferred body temperature. An alternative prediction is that lizards would develop a "behavioural fever" response to infection, where parasitised individuals increase their body temperature above the preferred value for non-parasitised individuals by seeking longer exposure to warmth. These two predictions are not mutually exclusive, and there is a possibility both can occur in response to parasite infection, resulting in higher preferred temperatures and but lower thermoregulation precision for parasitised individuals.

An alternate hypothesis is that the behaviour of parasitised lizards is not caused by the parasites but due to intra-specific variations of behavioural type in the wild (Horváth et al., 2024), established prior to contracting the parasite infection. This would make the behaviour shift not a consequence of parasitism, but a factor that can promote it. Two predictions could align with this hypothesis. First, lizards with naturally higher thermoregulatory precision would be more heavily infected, as parasites might benefit from the more stable thermal environment (Megía-Palma et al., 2020b, 2022a). In this case, we expect to detect a significant positive relationship between thermoregulation precision and parasite infection intensity, but no significant relationship with whether individuals are parasitised, as a more stable in-host thermal environment should only affect the parasite post-infection. Second, if no relationship is found between parasite infection and thermoregulation, it would instead suggest the previously obtained results stem from naturally bolder lizards being more exposed and susceptible to infection by parasites (Barrientos and Megía-Palma, 2021; Payne et al., 2024). All these hypotheses and predictions are summarized in Table 1.

Considering parasite influence over ectotherms beyond lizards, cases exist where parasites directly and actively manipulate their host's thermoregulatory behaviours to acquire fitness advantages. For example, parasitic flatworms can alter snail and fish thermal preferences to steer the host towards warmer environments which are beneficial for the parasite development or transmission (Bates et al., 2011; Macnab and Barber, 2011). However, to the best of our knowledge, no evidence currently supports such manipulative relationships between parasites and lizard hosts.

To assess whether blood parasite infection is associated with *P. lusitanicus* lizards' thermoregulation, and by which of the above-presented mechanisms, we recorded the temperature of 28 lizards at regular intervals throughout their natural daily thermoregulatory activity period in an experimental thermal gradient. We then analysed standard thermal traits (details in methods) in relation to lizard size, body condition and leucocyte counts, and parasite infection data (infection status and intensity). With this, we should also be able to identify whether these changes are plausibly linked to previously observed behavioural shifts in *P. lusitanicus* in response to parasitism (Faria et al., 2024). Furthermore, this should allow us to infer whether these shifts correspond to a host response to parasite infection, or an adaptive strategy of the blood parasites to increase their reproductive success in this host population. For the latter, if blood parasite infection impaired the lizard host's thermoregulatory precision by increasing their exploratory behaviour (Faria et al., 2024), the blood parasite would increase its own chances of sexual reproduction in haematophagous mite definitive hosts because infestation rates by these ectoparasites increase proportionally to lizard exploratory activity (Barrientos and Megía-Palma, 2021).

2. Material and methods

2.1. Ethical statement

Ethical approval was not required for this study because it involved only observational laboratory experiments on captive lizards and the collection of a small, non-lethal tail-tip tissue sample using established best-practice methods. The procedures did not involve invasive surgery or experimental manipulation beyond standard animal husbandry. All

Table 1

Summary of hypotheses, predictions, and respective principal expected and potentially exclusive significant relationships.

Hypothesis	Prediction	Expected significant relationships	Potentially exclusive significant relationships
Parasite infection impacts thermoregulatory behaviour	Parasite infection affects thermoregulation precision Parasite infection triggers behavioural fever	Thermoregulation precision ~ Parasite infection status (+) Preferred body temperature ~ Parasite infection status (+)	- -
Natural differences in behaviour promote parasite infection	Increased thermoregulation precision promotes parasite infection growth Increased boldness promotes exposure and infection	Thermoregulation precision ~ Parasite infection intensity (+) -	Thermoregulation precision ~ Parasite infection status (+) Any thermoregulation variable ~ Any parasite infection variable

experiments, including animal capture, housing, and manipulation, were authorized and followed the national guidelines from the *Instituto da Conservação da Natureza e das Florestas, I. P.* (ICNF), Portugal. These were officialised under the approval no. S-016562/2024, comprising the licenses no. 578/2024/CAPT, 579/2024/CAPT, and 12/2024/DET. All experiments were performed and reported according to the “Animal Research: Reporting of In Vivo Experiments (ARRIVE) guidelines 2.0”.

2.2. Study system and husbandry

We captured 28 adult male *Podarcis lusitanicus* by noosing, during June 2024, within the breeding season (Carretero et al., 2006). This population inhabits a semi-rural area next to the town of Moledo, Portugal (41°50'N 8°51'W), on the Atlantic coast, composed of a mosaic of small traditionally cultivated and abandoned plots surrounded by dry stone walls. It is separated from the sea by a walking trail and over 100 m of sandy dunes and rocky outcrops. This population of *P. lusitanicus* occurs in sympatry with two other lacertid species, *Podarcis bocagei* and *Timon lepidus*, although with microhabitat-usage differences. Both *Podarcis* species are known hosts of blood parasites from the suborders Adeleorina (*Karyolysus* sp.) and, less frequently, Eimeriorina (*Schellackia* sp.) (Faria et al., 2024; Maia et al., 2014).

Following capture, lizards were transported to the experimental facilities in individual cloth bags and housed in a room (ca. 8 by 5 m) with temperature set at 22 °C, controlled by the means of air conditioning, which is expected to also maintain stable relative humidity, although this was not measured. Lizards were placed in individual transparent acrylic terraria (23.0 × 15.5 × 17.0 cm), containing a movable ceramic shelter (10.0 × 10.0 cm) and a ceramic basking outcrop (diameter ca. 6.0–8.0 cm and height ca. 3.0 cm). Terraria were kept under a light cycle of 10.5 h between 08:30–19:00 h with both 46 W regular lamps and ultraviolet lamps. Lizards were provided with a single *Tenebrio molitor* larva once/day and water *ad libitum*. Lizards were left in the terraria to acclimatize and reduce post-capture stress for 3 days prior to the first experiment. The movable ceramic shelters allowed us to move lizards between their terrarium and the thermal gradient without touching them, by gently tipping the shelter box into the thermal gradient.

2.3. Experimental setup

The experiments took place in the same room where the lizards were housed. Each of the 10 thermal gradients consisted of a rectangular (40 × 150 × 40 cm), open-top opaque terrarium with fine gravel substrate, a 1.5 cm wide channel for water along one of the long sides, and an overhanging 46 W lightbulb 40 cm above the substrate closer to one of the ends of the lane, providing a gradient from 45 °C under the lightbulb to 20 °C on the farthest point from it (substrate temperature). Room temperature during the experiments was monitored constantly and recorded at every 30 min with a digital thermometer, with recorded values having a mean and standard deviation (SD) of 24.80 ± 0.70 °C. The increase from the standard room temperature of 22 °C maintained by the means of air conditioning was considered due to the lamps used in the thermoregulation experiments.

The experiments took place on three consecutive days, between 09:00–18:30 h (within lizard activity time in the field). Lizards were split across three batches (one batch for each day), and each lizard was tested only once. Lizards were individually placed on the thermal gradients at 19:00 h on the day prior to the experiment and left overnight to acclimatize. At 08:00 h on the day of the experiment, lights over the thermal gradients were turned on and the water channel carefully refilled with a nozzle bottle at the furthest point possible from the lizard. Lizard body temperature was recorded every 30 min, from 09:00–18:30 p.m., using a handheld infrared (IR) camera (model Flir T335) at ca. 30 cm above the lizard. Two thermal photographs were obtained at each time point, using FLIR Tools v6.4 (Teledyne FLIR, 2018) to determine the temperature on the top of the tail base of the lizards, and the average of these values was used for statistical analyses. We chose the tail base to record temperature because the IR camera temperature measurements from this region are strongly correlated with cloacal temperature (Barroso et al., 2016).

To minimize and standardise the effects from human presence in the room, the same person (JFF) recorded the thermal photographs during the three days of the experiment. On each day, for the duration of the experiment, they remained sitting in the room, on the opposite side from the gradients, and approached only to record the thermal photographs. The thermal gradients were approached carefully to not startle the lizards, which did not run away from their current spot while the photographs were being taken, although they did, on occasion, tilt their head towards the camera, indicating they were aware of human presence in those moments.

After each experiment, we mixed the thermal gradient substrate to create a uniform chemical gradient (Duvall, 1979; Megía-Palma et al., 2024b), before placing the next batch of lizards in the thermal gradients to stay overnight.

We then weighed the lizards with a digital scale to the nearest 0.1 g and measured the snout-vent length (SVL), from the tip of the snout to the posterior point of the cloacal shield, using callipers to the nearest 0.1 mm. All lizards had complete tails, and tail condition was assessed as original or regenerated. We collected a blood smear from a small incision on the tail tip. Blood smears were photographed on an optical microscope with an Olympus EP50 digital camera at an amplification of 400x, and at least 3000 erythrocytes per sample were screened for blood parasites with ImageJ v1.54 (Schneider et al., 2012), using morphological traits to identify parasite genus (Blázquez-Castro et al., 2023; Faria et al., 2024). Infection status (parasitised or non-parasitised) was determined by the presence of parasites in the sample (minimum of 1 parasite in the sample). We also recorded infection intensity as the number of infected erythrocytes per 3000 counted erythrocytes. Total leucocytes were also counted alongside erythrocytes, to obtain a proxy for immunocompetence representing cell-mediated surveillance and response (Bílková et al., 2015; Zhang and Zhao, 2015; Zimmerman et al., 2010). For the statistical analysis, we quantified the number of leucocytes present in the smears per 3000 erythrocytes.

2.4. Data analysis

All analyses were performed using R software v4.4.0 (R Core Team, 2024). We calculated body condition (BC) from the residuals of a linear regression of the decimal logarithm of body mass on the decimal logarithm of SVL (Warner et al., 2016). We considered recorded maximum temperatures ($T_{\max}$) as the highest recorded body temperature reached by each lizard during the experiment. To calculate preferred body temperature (T_p) and thermoregulatory precision (T_{SD}), we discarded initial measurements of each lizard until their first maximum (Jiménez-Robles and De la Riva, 2019) and excluded extreme outliers from the temperature measurements, defined as values below the 1st quartile and above the 3rd quartile by more than 1.5 times the inter-quartile range. With the resulting dataset, T_p was calculated as the median of the recorded temperatures (Pough and Gans, 1982; Thompson et al., 2023) and T_{SD} as the standard deviation of the same temperatures (Paranjpe et al., 2014), meaning that higher T_{SD} values correspond to lower thermoregulatory precision.

To analyse the effect of the host traits on parasite infection parameters of status and intensity, we used Generalized Linear Models (GLMs) with, respectively, binomial and log-linked Gamma distributions, and considered the predictor variables SVL, BC, tail condition and leucocyte counts. Only parasitised individuals were used in the model for parasite infection intensity. To analyse the effects of host traits and parasite infection on the thermal variables $T_{\max}$, T_p , and T_{SD} , we used Linear Models (LMs), considering the predictor variables SVL, BC, tail condition, leucocyte counts, parasite infection status and intensity. We considered using LMMs for the thermal variables, including day of the experiment as a random factor, as the lizards were tested in three different batches across three days and ambient conditions may have differed slightly; however, the variance for that random effect was estimated at zero, indicating models both with or without batch as a random factor gave similar results and the same conclusions. To analyse the time series records of lizard body temperature throughout the day, we used Linear Mixed Models (LMMs) considering the same predictor variables as for the other thermal variables plus the interactions between time of day (as a continuous variable) and both parasite infection status (parasitised/non-parasitised) and intensity and individual lizard as a random factor. To analyse the effect of parasite infection and variation of thermoregulatory behaviours (using the thermal variables as proxies) on the cellular immune response of lizards, another log linked Gamma GLM was built with the latter as response variables and considering the predictor variables SVL, BC, tail condition, parasite infection status and intensity and the thermal variables ($T_{\max}$, T_p , and T_{SD}), as well as the interactions between both parasite infection parameters and each of the thermal variables.

We built a global model for each analysis, including all considered predictor variables described before and used an adjusted Variance Inflation Factor (VIF) function (package: *performance*; Lüdtke et al., 2021) to check for multicollinearity within the global models without variable interactions (which are not supported by VIF), iteratively dropping the highest scoring variable until all VIF scores were under a maximum threshold of 5 (James et al., 2023; Neter et al., 1996). We implemented a multimodel inference approach using the function *dredge* in the R package MuMIn (Bartoń, 2026) with an information criterion corrected for small sample sizes (AICc; Bedrick and Tsai, 1994) to identify the most informative models (with AICc scores differing by less than two units from the best scoring model). We then used the conditional model-average coefficients to calculate the relative importance of each predictor variable in the final average model and retrieve the estimated significance ($\alpha < 0.05$) of the effects and their z-standardised β coefficient (Estimate) $\pm$ adjusted standard error, to identify the effect direction and strength. We finally tested the most informative models' simulated residuals for uniformity, quantile deviation, overdispersion, and zero-inflation (package: *DHARMa*; Hartig et al., 2024).

3. Results

Eight of 28 (29% prevalence) *P. lusitanicus* were infected with *Karyolysus* sp. (Adeleorina). Intensity of infection was 1–6 parasites per 3000 erythrocytes (mean $\pm$ SD = 3.09 ± 2.25), corresponding to 0.1% infected erythrocytes on average. Parasite prevalence and intensity were similar to the assessment obtained for this population in 2022, but this time no *Schellackia* parasites were detected (Faria et al., 2024). Lizard body traits SVL, BC, tail condition, and leucocyte counts were not correlated, nor were they significantly related to parasite infection variables (status and intensity) in their respective global models (max. VIF = 1.99 for status and max. VIF = 3.45 for intensity). Only the 8 parasitised individuals were considered in the analyses of parasite infection intensity.

Median preferred temperature (T_p) was 37.35 °C (SE = 0.14) for non-parasitised males of *P. lusitanicus* and 38.10 °C (SE = 0.32) for parasitised lizards (Fig. 1A). For the global model, VIF analysis indicated only low correlations (max. VIF = 4.16 for parasite infection intensity). None of the top models included parasite infection parameters as predictor variables. Interindividual differences in T_p were significantly related to body condition with lizards with higher body condition selecting higher temperatures independently of parasitism (Fig. 1B–Table 2).

For the global model for thermoregulatory precision (T_{SD}), max. VIF = 4.16 for parasite infection intensity. The response variable was significantly related with parasite infection status and host tail condition (Table 2). Parasitised lizards had lower thermoregulatory precision (higher T_{SD} values; Fig. 2A and Table 2). Lizards with regenerated tails exhibited higher thermoregulatory precision than ones with original tails (Table 2), despite the same median value of T_{SD} for the two groups (Fig. 2A).

For the global model for the recorded maximum temperatures ($T_{\max}$), max. VIF = 4.16 for parasite infection intensity. The null model was one of the two best scoring models, with the other including only parasite infection status (importance = 0.28), indicating none of the predictor variables provided a strong explanation for the observed variation in $T_{\max}$. Nevertheless, graphical analysis of the values suggests a non-significant trend towards a higher recorded maximum in parasitised lizards (Fig. 2B).

For the time series records of lizard body temperature throughout the day, the global model max. VIF = 4.15 for parasite infection intensity. Lizard body temperature was significantly related to time of day, but not with any parasite infection parameter (Table 2). The interaction between time of day and infection status did not appear in any of the top models, despite an apparent graphical trend suggesting that the degree of variation of body temperature across the day is different for parasitised and non-parasitised individuals (Fig. 3).

For the global model for leucocyte counts, max. VIF = 4.75 for parasite infection status. While no significant relationships were identified with any predictor variables by themselves, all the interactions between the parasite infection parameters and each of the thermal variables (T_p , $T_{\max}$, T_{SD}) were significantly related with leucocyte counts, albeit at low importance (importance < 0.37; Table 2). This indicates that the effect of the thermal variables over leucocyte counts varied with whether the lizards were parasitised or not and with the intensity of the parasite infection. While the general relationships between the thermal variables (T_p , $T_{\max}$, T_{SD}) and leucocyte counts trended towards increasing leucocyte counts with higher T_p and $T_{\max}$ and lower T_{SD} (the latter indicating greater thermoregulatory precision), the magnitude of this increase in leucocyte counts in response to the thermal variables was significantly reduced for parasitised individuals and for those with higher intensity of infection (Table 2).

4. Discussion

Parasitised lizards exhibited significantly less precise

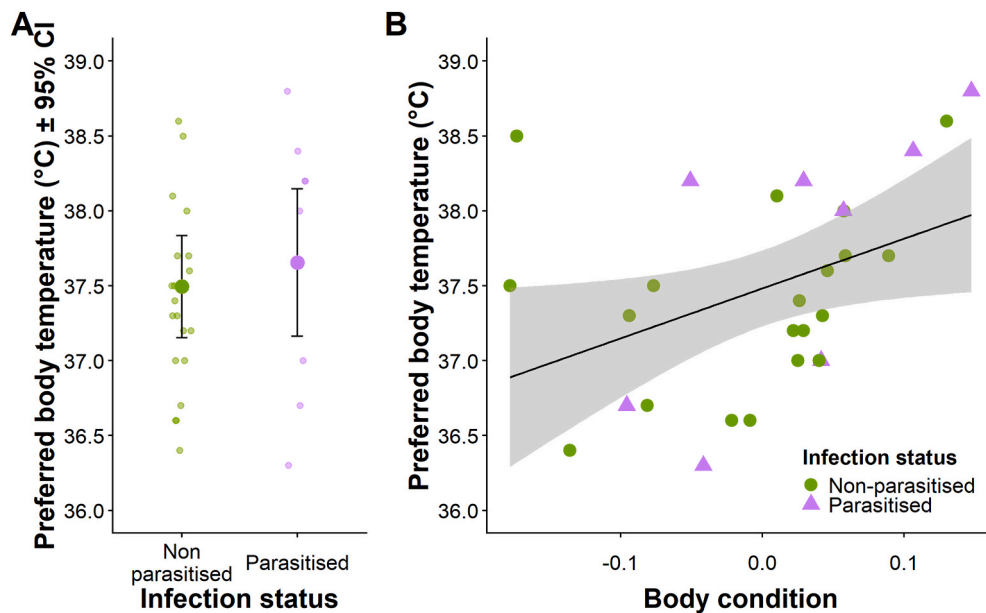


Fig. 1. Body condition, but not infection status, can explain variation in preferred body temperature. (A) Relationship between predicted preferred body temperature (T_p ; 95% confidence interval) and parasite infection status, non-significant. Raw data points underneath. (B) Relationship between preferred body temperature (T_p ; 95% confidence interval in gray) and body condition, significant. Triangles were used for parasitised individuals to permit identification of overlapping points.

thermoregulatory behaviour (i.e., higher SD of body temperatures) (Fig. 3). This illustrates how blood parasites can be associated with changes to thermal physiology, which is fundamental to life history traits in ectotherms, in contrast to most studies which focus on health markers. Additionally, the simultaneous significant effect of parasite infection status (parasitised or non-parasitised) and tail condition in thermoregulatory precision further illustrates how important it is to take an integrative approach. The cellular immune response of parasitised lizards also developed less from increases in body temperature and thermoregulatory precision, suggesting that parasitic infection may constrain the mounting of such defences.

Reduced thermoregulatory precision has been documented in another lacertid lizard: *Iberolacerta cyreni* (Horváth et al., 2020), in which, like *P. lusitanicus*, parasitised individuals had lower thermoregulatory precision than non-parasitised lizards. However, contrary to previous studies on *I. cyreni* (Horváth et al., 2020; Megía-Palma et al., 2020b), it was parasite infection status, and not intensity, that had a significant effect on the thermoregulatory precision of *P. lusitanicus*. This might be explained by differences in the way parasitism was assessed. As in the case of *I. cyreni* (Horváth et al., 2020), higher erythrocyte counts (12,000 per individual) possibly provided increased variability in parasite intensity (mean = 18, SD = 25.87) values and reduced the chance of false negatives. However, previous studies of the same *P. lusitanicus* population as this study, produced similar patterns in intensity (Maia, 2015; Vasconcelos, 2021), and the occurrence of false negatives is very rare when compared to PCR detection methods (Maia et al., 2014). Most likely, the difference in significant parasite infection variables between the two species (infection status for *P. lusitanicus* vs. infection intensity for *I. cyreni*) results from different host-parasite relationships and coevolution histories. The mountain populations of *I. cyreni* exist at higher population densities, facilitating vector exposure and parasite transmission (Amo et al., 2004; Godfrey et al., 2009), cf. (Sorci et al., 1997), and are exposed to unpredictable weather and longer cold periods, impairing their immune systems (Johnson et al., 2019). These factors likely contribute to the higher prevalence (around 89% and 95%) and intensity of blood parasites in *I. cyreni* (Horváth et al., 2020; Megía-Palma et al., 2020b), thereby allowing the development of varying levels of severity for the impact of parasite

infection upon the lizard hosts. Maintaining thermoregulatory precision can be a costly endeavour for lizards, often requiring a trade-off between heat seeking and safety or foraging (Paranjpe et al., 2014; Sartorius et al., 2002). As our results show a negative impact of parasite infection on thermoregulatory precision, with parasitised lizards having higher difficulty maintaining a stable temperature throughout the day, this might indicate a lack of available energy to allocate to thermoregulatory behaviours, thus suggesting the parasite infection imposes extra energy costs. This higher cost, might also be associated with the lack of an increased cell-mediated immune response in parasitised individuals, as no significant relationships between leucocyte counts and either parasite infection or the thermoregulatory variables (T_p , T_{SD} , T_{max}) were observed, which is consistent with other studies on lacertids (Damas-Moreira et al., 2022). Considering blood parasite infections tend to be chronic conditions (Megía-Palma et al., 2023), it is possible that once the parasite infection is established, lizard host immune responses towards the infection would subside to minimize the physiological costs that would come with such prolonged immune activity, especially in short-lived lizards such as these, where enduring the infection might be the least costly strategy (Lee et al., 2023). Furthermore, *Karyolysus* sp. is typically observed within a parasitophorous vacuole in the cytoplasm of the host's infected erythrocytes, which might delay detection and response by cellular immune means (Megía-Palma et al., 2023) to a point where it might not even trigger a response in short-lived host organisms such as these (Lee et al., 2023), which could explain the lack of an increase in host leucocyte density in parasitised lizards. However, while we did not identify a direct significant relationship between leucocyte density and either parasite infection or thermal variables, the interactions between these two groups of variables were significant. While non-significant, higher T_p and T_{max} and more precise thermoregulation all showed a trend for a positive relation with leucocyte counts; however, in parasitised lizards, these effects were significantly less. This would suggest that, while the parasitic infection or the thermoregulatory precision changes are not significantly reducing the baseline levels of cellular immune response in these hosts, the parasite infection does seem to hinder the possible positive effects of certain thermoregulatory behaviours for this form of immune response. Therefore, it might be that these lizards are not simply enduring the

Table 2

Results of model averaging for the thermal variables T_p , T_{max} , and T_{SD} , and the time series records of lizard temperature across time, indicating the number of models within the selection threshold for creating the final averaged model ($AIC_c \leq 2$). Significant relationships ($p \leq 0.05$) are marked in bold. SVL = snout-vent length. Adj SE = adjusted standard error.

Median preferred temperature (T_p) (2 models)		Importance	Estimate	Adj. SE	z-value	P-value
(Intercept)			37.390	0.247	6.506	<0.001
Body condition	1.00	3.457	1.579	2.190	0.029	
Tail condition	0.32	0.351	0.348	1.007	0.314	
Recorded maximum temperature (T_{max}) (2 models)		Importance	Estimate	Adj. SE	z-value	P-value
(Intercept)			40.235	0.195	206.310	<0.001
Parasite infection status (parasitised)	0.28	0.288	0.405	0.710	0.478	
Thermoregulatory precision (T_{SD}) (3 models)		Importance	Estimate	Adj. SE	z-value	P-value
(Intercept)			2.138	0.205	10.421	<0.001
Body condition	0.46	-1.661	0.983	1.690	0.091	
Tail condition (regenerated)	1.00	-0.484	0.216	2.243	0.025	
Parasite infection status (parasitised)	1.00	0.399	0.182	2.187	0.029	
Leucocyte counts	0.18	0.003	0.003	1.110	0.267	
Time series records of lizard temperature (2 models)		Importance	Estimate	Adj. SE	z-value	P-value
(Intercept)			37.951	0.293	129.473	<0.001
Body condition	1.00	2.506	1.479	1.694	0.090	
Tail condition (regenerated)	0.35	0.359	0.325	1.102	0.270	
Time of day	1.00	-0.071	0.015	4.789	<	0.001
Leucocyte counts (6 models)		Importance	Estimate	Adj. SE	z-value	P-value
(Intercept)			-0.876	6.748	0.130	0.897
Parasite infection status (parasitised)	0.59	22.638	13.832	1.637	0.102	
P. infection status (parasitised): T_p	0.18	-0.707	0.281	2.519	0.012	
P. infection status (parasitised): T_{max}	0.33	-0.806	0.260	3.103	0.002	
P. infection status (parasitised): T_{SD}	0.33	1.685	0.616	2.735	0.006	
Parasite infection intensity	0.76	4.271	3.651	1.170	0.242	
Parasite infection intensity: T_p	0.37	-0.158	0.070	2.261	0.024	
Parasite infection intensity: T_{max}	0.12	-0.179	0.071	2.510	0.012	
Parasite infection intensity: T_{SD}	0.12	0.416	0.161	2.580	0.010	
T_p	0.55	0.198	0.186	1.065	0.287	
T_{max}	0.45	0.037	0.121	0.309	0.758	
T_{SD}	0.45	-0.064	0.258	0.247	0.805	

parasite infections to save energy (cf. Lee et al., 2023), but instead that the parasitic infection itself might be disrupting some of the mechanisms that allow these organisms to develop responses beyond baseline leucocyte levels.

The observed positive relationship between thermoregulation precision and parasite infection, together with the positive relationship between boldness and parasite infection previously observed in the same

host species (Faria et al., 2024), suggests that the effects of parasitic infection can potentially cascade into multiple changes in host behaviour. The increased tendency towards exposure found in a previous experiment (Faria et al., 2024) could suggest parasitised *P. lusitanicus* sought more basking time and more direct light exposure in order to attain higher body temperatures, consistent with behavioural fever; however, we found no evidence of this phenomenon in our study.

Both preferred body temperature and the recorded maximum were nearly the same for parasitised and non-parasitised individuals. This again is in line with the results of (Horváth et al., 2020) and congruent with other lizard studies where parasitised individuals did not exhibit an increase in preferred body temperature (Lee et al., 2023; Megía-Palma et al., 2020b; Schall, 1990). An alternative hypothesis that animals could increase sunlight exposure to combat parasite infection through ultraviolet radiation or desiccation, and not by increasing body temperature, has been proposed for birds (Bush and Clayton, 2018); however, this was in response to ectoparasites. The behavioural shift in parasitised lizards towards increased exposure might then be seen as an example of the asset protection principle, prompting individuals of lower reproductive value to be more risk-prone — a response to parasitism that had already been identified in other lizard host-parasite systems (Horváth et al., 2016; Sorci et al., 1996). However, it could also result from parasitised lizards behaving more erratically due to a putative parasite-imposed impairment of the lizard host self-thermal perception. To address this question, we would need to analyse different risk-taking parameters or to perform exploratory experiments without the “light exposure variable”, for example, by performing the tests in a heated floor arena (Lee et al., 2023), to understand whether the increased willingness to leave a refuge and expose themselves to parasitised individuals really is to compensate for their lower reproductive value with increased risk taking or an attempt to meet their thermoregulation requirements against the parasite-imposed lower thermoregulatory precision. Another approach could be to pharmacologically treat lizards with the parasite infection (Foronda et al., 2007) and analyse whether the behaviour of the same individuals would change pre- and post-treatment, although this approach was ineffective for treating Adeleorine parasites in other small lacertids (Megía-Palma et al., 2020b).

With the lower thermoregulatory precision being of no apparent direct benefit to *P. lusitanicus*, it could instead be induced by the parasite to weaken its host and obtain some fitness advantage. However, our results provide no evidence that lowering the host's thermoregulatory precision is directly advantageous for these parasites, to the contrary, the lack of a stable thermal environment might even constrain parasite fitness (Oppliger et al., 1996) and be one of the reasons why infection intensity remains at low levels for this population. Nevertheless, it could still be significant for the life-cycle of *Karyolysus* sp. beyond its infective stage in the lizard host. It is in ectoparasite definitive hosts (typically haematophagous mites of the order Mesostigmata) that *Karyolysus* sp. maximizes its fitness by reproducing sexually and thus increasing its genetic diversity (Svahn, 1975; Telford, 2009). By inducing an alteration in lizard thermoregulatory behaviour, the parasite may increase its chances of transmission to its final host by increasing its lizard host's exploratory behaviour in response to lower thermoregulatory precision (Barrientos and Megía-Palma, 2021). Furthermore, by inducing this change without altering the lizard host preferred temperature (as we observed by the lack of T_p variation with parasite infection), the parasite would be maintaining its viability (and that of the lizard) while infecting this host, despite the thermal instability caused by the disruption of the lizard thermoregulatory precision.

We found no support for the alternative hypothesis that the observed increased boldness found in a previous study (Faria et al., 2024) could precede parasite infection, by being an inherent behaviour type of specific individuals that would increase the chances of contracting these parasites. If that was the case, we would expect more uninfected lizards to have a low thermoregulatory precision, eventually resulting in

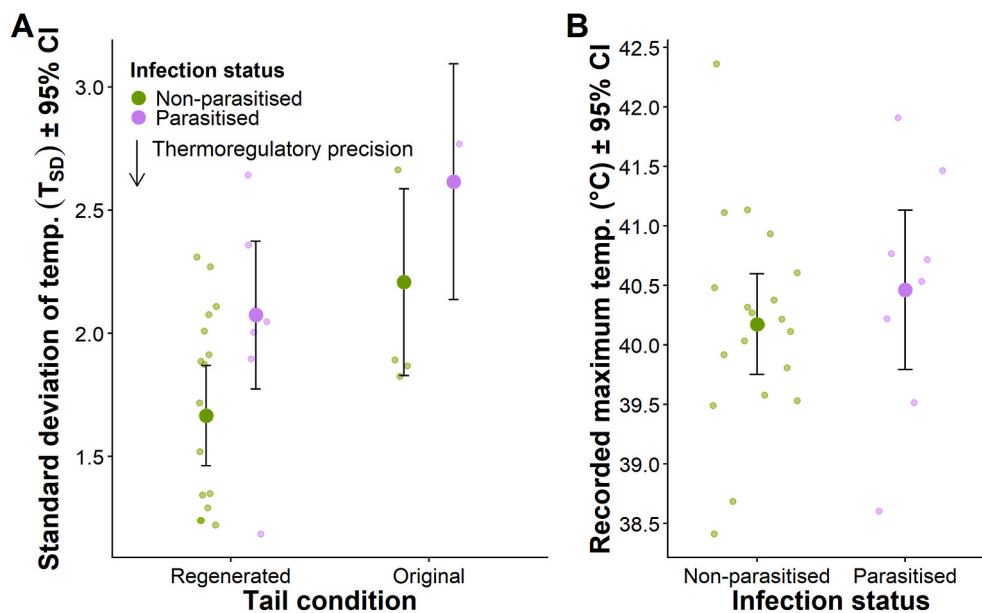


Fig. 2. Infection status and tail condition can explain thermoregulatory precision, but recorded maximum temperature is independent of parasitism. (A) Relationship between predicted thermoregulatory precision scores as standard deviation of lizard body temperatures (T_{SD} ; 95% confidence interval) and parasite infection status according to tail condition. The arrow represents the direction of increased thermoregulatory precision, as lower T_{SD} values equal higher precision. Significant effects were identified for both parasite infection status and tail condition. Raw data points underneath. (B) Relationship between predicted maximum temperature (T_{max} ; 95% confidence interval) and infection status, non-significant. Raw data points underneath.

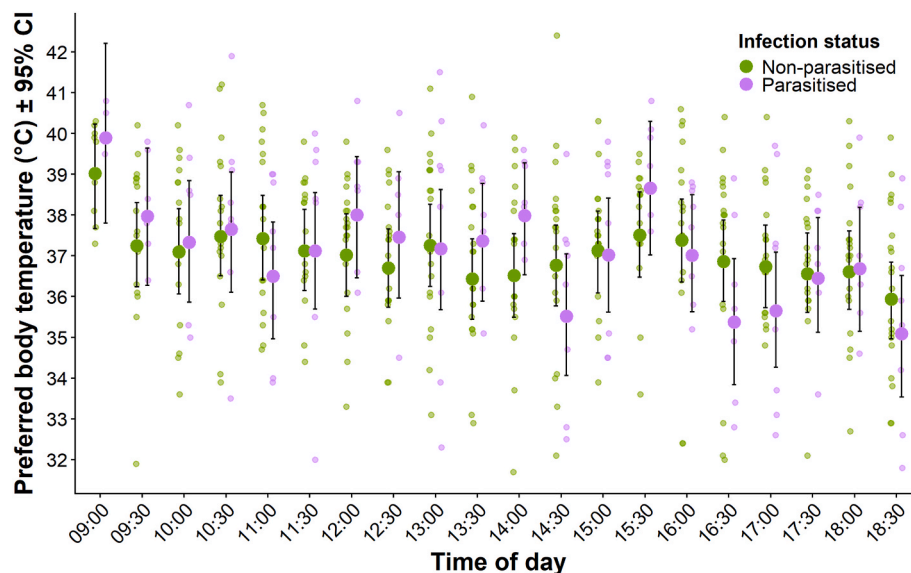


Fig. 3. Lizard body temperature during experiment varies differently for parasitised individuals along the day. Predicted lizard body temperature (including 95% confidence interval) throughout the day for parasitised and non-parasitised individuals. The presented times correspond to the 20 temperature recording moments during the experiment. Raw data points underneath.

non-significant differences between parasitised and non-parasitised lizards since the prevalence of infection in this population is relatively low (29% in this study). The lack of a significant relation between lower thermoregulatory precision and increased parasite infection intensity or reduced leucocyte density also makes it unlikely that a behaviour-type linked to naturally lower thermoregulation precision would be providing a benefit for this parasite infection to thrive post-initial infection in such hosts and explain the observed variation in thermoregulatory precision.

Tail autotomy and the subsequent regenerative process have long been hypothesised to impact thermoregulatory behaviour in lizards (Martin and Salvador, 1993). The hypotheses have varied on whether

tail loss would lead to improved thermoregulatory behaviours, as these lizards should try to optimize thermal conditions for regeneration, or worsen them as lizards without complete tails would suffer from mobility issues and shift energy costs to regeneration instead of searching for optimal thermal conditions. While our sampling included only lizards with complete tails, we have distinguished between those with original tails versus regenerated ones and found that lizards with regenerated tails had significantly higher thermoregulatory precision than those with original tails. This suggests lizards with regenerated tails were actively trying to maintain a more precise thermoregulation, even when the tail appears already fully regrown. This would further suggest that regenerated tails, even when fully regrown, might be

physiologically different from an original tail, carrying the need for different thermoregulatory behaviours further into the lizard's lifespan. Contrarily to our results, previous studies have not identified significant relationships between tail loss and thermal preferences, heating rates, or microhabitat usage (Fernández-Rodríguez et al., 2021; Herczeg et al., 2004; Martin and Salvador, 1993; Zamora-Camacho et al., 2015), despite one identifying a significant difference in basking time (Martin and Salvador, 1993). Most of these previous studies, however, relied on a single temperature measurement and none assessed thermoregulatory precision, despite some mentioning it (Herczeg et al., 2004; Martin and Salvador, 1993). Furthermore, they have also grouped lizards as tailed or tailless, rather than considering those with regenerated tails as a separate group within tailed. Our results highlight the need to revisit research on the relationships between tail condition and thermoregulation, given the complexity of interactions in physiological processes and showcasing the need for time-series assessments instead of single-time measurements when studying behaviours and physiology.

In summary, we observed a significant change in thermoregulation-related parameters in parasitised *P. lusitanicus*, indicating the presence of Adeleorine blood parasites has a negative effect on the thermoregulatory precision of this lizard species. These changes are likely associated with an increased willingness to leave a refuge to access a heat source that had been previously identified (Faria et al., 2024); however, we cannot fully confirm this as the experiments were performed on different animals, albeit from the same populations, demographic group, and captured during the same season in different years. We could not ascertain if the loss of thermoregulatory precision is either a host response to infection or a parasite strategy to obtain a fitness advantage, at least during this life-stage of the parasite. Further, considering that we observed no significant direct fitness advantage to either host or parasite from lowering thermoregulation precision, we also cannot rule out the possibility that this impact of *Karyolysus* on *P. lusitanicus* could be a by-product of the infection, and not an adaptative strategy from either of the parties in response to the other. This would suggest an incipient co-evolution story between this host-parasite pair (Megía-Palma et al., 2022b, 2023, 2024a). Nevertheless, we did identify that parasite infection significantly disrupted the relationship between thermoregulation and cellular immune response, constraining possible benefits that modulating thermoregulatory behaviours could have on mounting this form of immune response. The identification of a significant direct negative effect of blood parasites in a vital trait (thermoregulation precision) and of the significant negative effects on the immune response by the interactions between non-significant factors (all three thermal variables with both parasite infection parameters) on a lizard species for which no negative impacts of parasite infection had been previously found highlights both how broad and complex the impacts of these parasites can be, and the importance of analysing other, less commonly used, indicators of host health in more comprehensive studies.

Data statement

The data collected and used in this research work has been shared as supplementary material, under file “S1 Data.csv”.

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CRediT authorship contribution statement

J. Filipe Faria: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Software, Visualization, Writing –

original draft, Writing – review & editing. **Rodrigo Megía-Palma:** Conceptualization, Supervision, Validation, Writing – review & editing. **Martin J. Whiting:** Conceptualization, Supervision, Validation, Writing – review & editing. **D. James Harris:** Conceptualization, Funding acquisition, Methodology, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Joaquim Filipe Faria reports financial support was provided by Fundação para a Ciência e Tecnologia (FCT), Portugal. D. James Harris reports financial support was provided by Fundação para a Ciência e Tecnologia (FCT), Portugal. Rodrigo Megía-Palma reports financial support was provided by Fundação para a Ciência e Tecnologia (FCT), Portugal. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jtherbio.2026.104514>.

Data availability

The data has been shared as supplementary material.

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